Automotive Cooling System



Category: **Automotive** Type: **Full SOP** Priority: **Medium** 2 labs

Covers these lab items: Automotive Cooling System; Cooling system

What it is used for

An automotive cooling system removes excess engine heat and maintains the engine within its designed operating temperature range. Correct inspection and servicing help prevent overheating, coolant loss, and engine damage.

Main parts

- Radiator and cooling fan
- Water pump and drive belt or electric drive
- Thermostat and coolant passages
- Coolant hoses, clamps, and expansion tank
- Pressure cap and temperature sensor
- Heater core and control components where fitted

Before you start

1. Park the vehicle on level ground, apply the parking brake, and switch off the engine.
2. Allow the engine and cooling system to cool completely before opening the pressure cap.
3. Check the coolant level in the expansion tank and use the specified coolant type.
4. Inspect the radiator, hoses, clamps, water pump, drive belt, and tank for leaks or damage.
5. Check the cooling fan area for obstructions and inspect the belt or electrical connections as applicable.
6. Use the vehicle service manual to identify the correct pressure-test and filling procedures.

How to use

1. Inspect the system visually and identify signs of coolant leakage, corrosion, or damaged components.
2. Check the coolant level only at the designated reservoir when the engine is cold.
3. Pressure-test the system and cap only with approved equipment and the specified test pressure.
4. Repair or replace faulty hoses, clamps, seals, or components according to the service procedure.
5. Drain old coolant into an approved container when the procedure requires replacement.
6. Refill with the specified coolant mixture using the correct filling or bleeding method.
7. Start the engine only after the system is correctly filled and all tools are removed.
8. Monitor engine temperature and check for leaks, fan operation, and proper coolant circulation.
9. Allow the engine to cool and recheck the coolant level according to the service instructions.
10. Dispose of used coolant safely, clean the work area, and record the service performed.

⚠ Safety

- Never remove a pressure cap while the engine or coolant is hot.
- Keep hands, clothing, and tools away from cooling fans because they may start automatically.
- Wear suitable eye and hand protection when handling coolant and pressurized test equipment.
- Use the specified coolant and never mix incompatible coolant types.
- Collect coolant in a closed, labelled container and prevent spills onto soil or drains.
- Keep coolant away from children and animals because some formulations are toxic.

Care & storage

- Check coolant level and condition at the vehicle manufacturer's recommended intervals.
- Inspect hoses, clamps, radiator fins, pump, belt, and cap for wear or corrosion.
- Clean external radiator surfaces carefully without bending the cooling fins.
- Replace coolant and service the system according to the vehicle maintenance schedule.

Common mistakes to avoid

- Opening the pressure cap while hot can cause severe scalding.
- Using plain water or incompatible coolant can reduce corrosion protection and cooling performance.
- Trapped air after refilling can cause overheating or unstable temperature readings.
- Ignoring small leaks or a faulty pressure cap can lead to coolant loss and engine damage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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